

**Time:** 5 minutes**Outcome:** Interpret Competitive Efficiency & Model Limits.**Difficulty:** ●●○ Intermediate

Competitive Efficiency & Model Limits



THE CORE IDEA

Perfect competition is allocatively efficient when price equals marginal cost and productively efficient when firms produce at minimum average total cost. The benchmark requires private and social marginal costs to align. Apply the exact efficiency conditions and distinguish behavioral assumptions from welfare assumptions. Minimum average total cost represents the least-cost scale for producing that output in the benchmark.



HOW TO RECOGNIZE IT

- Apply the exact efficiency conditions and distinguish behavioral assumptions from welfare assumptions.
- Minimum average total cost represents the least-cost scale for producing that output in the benchmark.
- A representative competitive firm is in long-run equilibrium at the minimum point of ATC.
- Price reflects marginal willingness to pay while MC reflects marginal opportunity cost.

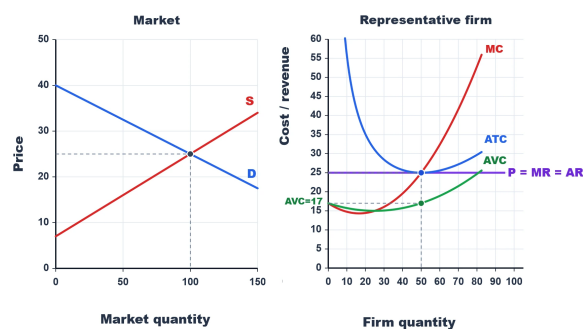


WATCH OUT

A common mistake is to equate efficiency with equity. Apply the exact efficiency conditions and distinguish behavioral assumptions from welfare assumptions. Minimum average total cost represents the least-cost scale for producing that output in the benchmark.



WORKED EXAMPLE: TESTING COMPETITIVE EFFICIENCY



At the long-run competitive outcome, the firm produces 50 units where price equals marginal cost and price also equals minimum ATC at \$25. $P = MC$ gives allocative efficiency, while production at minimum ATC gives productive efficiency. These conclusions depend on the competitive model's assumptions.



CHECK YOURSELF

$P=MC$ and $P>\min ATC$. Which interpretation is strongest?

**READY?**

Return to the game and master this concept.